SIGN-LANGUAGE

Spring 2026

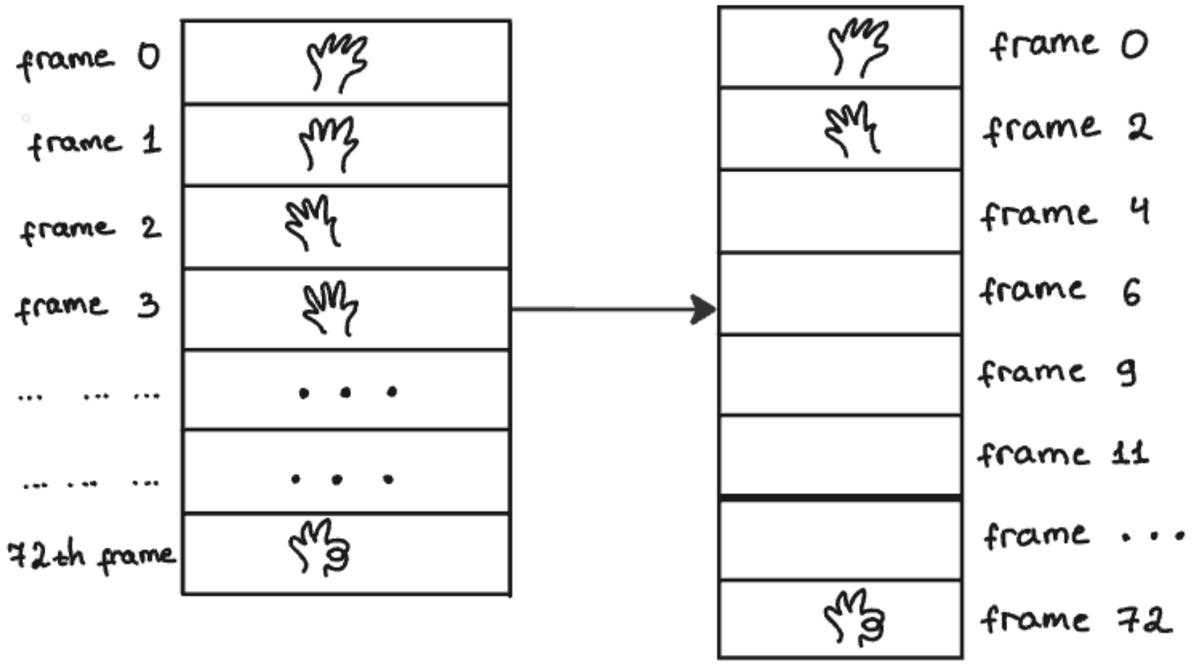
Meeting No. 5.5

System Development

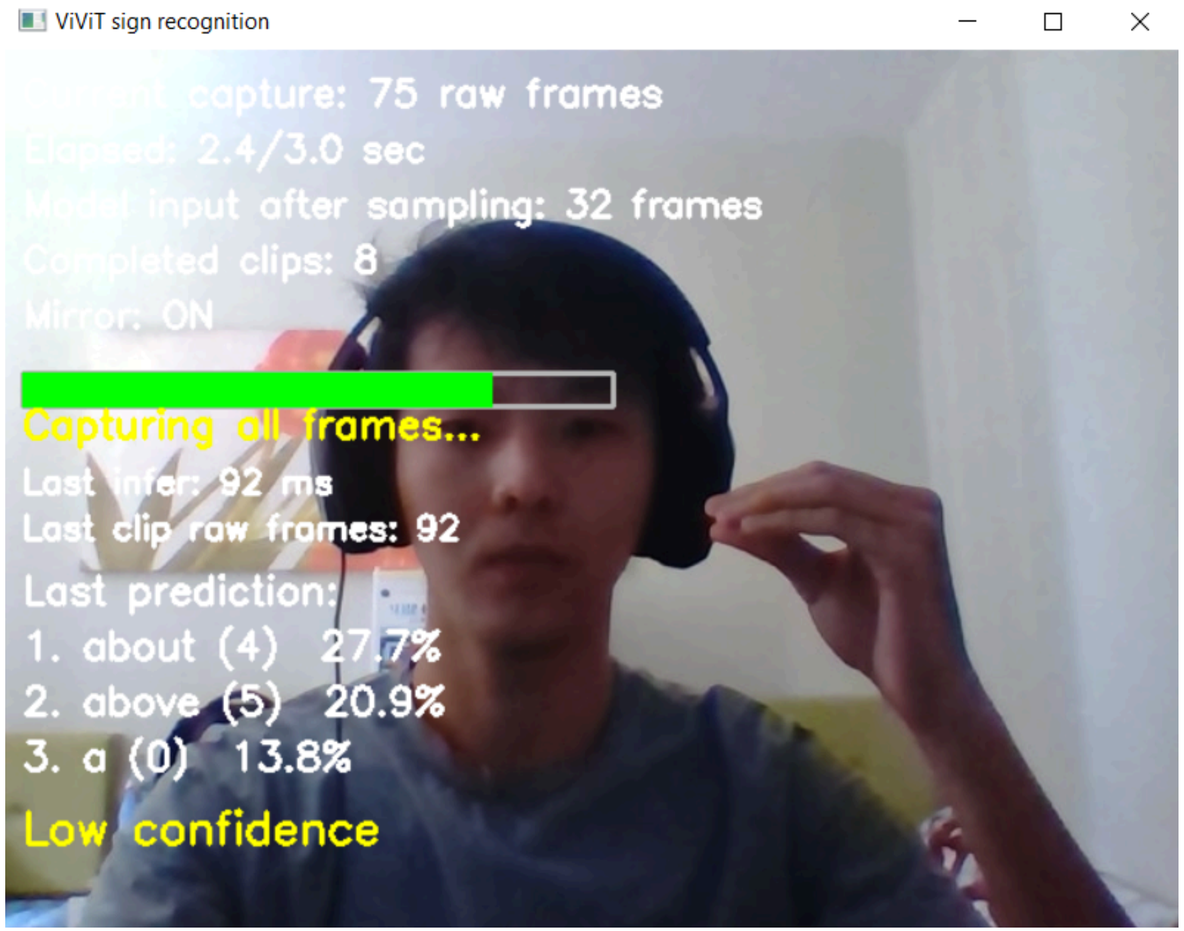
& Datasets

SYSTEM UPDATES

Firstly, raw frames will be captured
Uniformly distributed sampling was used
In total, 32 frames will be captured to predict
Same concept with the top-3 best predictions



SYSTEM UPDATES



ViViT Frame Sampling

If num_frames < 32:
Evenly repeat the frames starting from the first one until there are 32 frames.

Example:
num_frames = 30
frames = [1, 2, 3, ..., 30]
sampled frames = [1, 1, 2, 2, 3, 4, ..., 30]

If num_frames > 32:
Determine the step size (N) and sample every Nth frame until there are 32 frames.

Example:
num_frames = 70
frames = [1, 2, 3, ..., 70]
step size (N) = 70 // 32 = 2
sampled frames = [1, 3, 5, ..., 63]

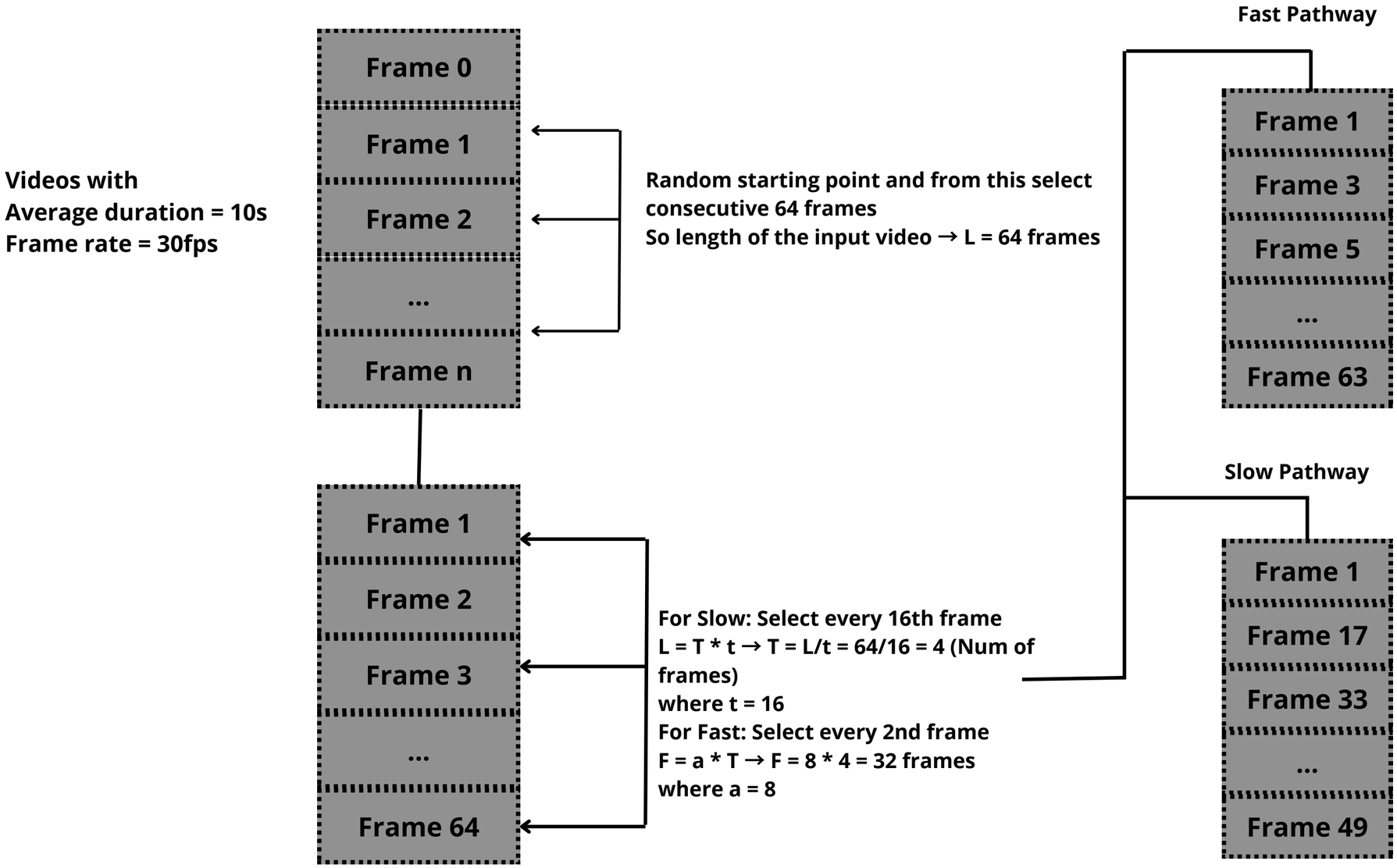
ViViT Frame Sampling

```python
def _sample_frames(self, frames):
    total = len(frames)

    if total >= self.num_frames:
        step = total // self.num_frames
        idx = [i * step for i in range(self.num_frames)]
    else:
        idx = np.linspace(0, total - 1, self.num_frames).astype(int)

    return [frames[i] for i in idx]
```

SlowFast on Kinetics in Research Paper



SlowFast frames sampling

```python
def _get_frames(self, video_path):
    cap = cv2.VideoCapture(video_path)
    total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
    window_size = 64

    if self.mode == 'train':
        start_frame = np.random.randint(0, max(1, total_frames - window_size))
    else:
        start_frame = max(0, (total_frames - window_size) // 2)
```

```python
def __getitem__(self, idx):
    video_path, label = self.samples[idx]
    raw_clip = self._get_frames(video_path)

    raw_clip = raw_clip.astype(np.float32) / 255.0
    raw_clip = (raw_clip - 0.45) / 0.225

    slow_stride = 8
    slow_idx = np.arange(8) * slow_stride
    slow = torch.from_numpy(raw_clip[slow_idx]).permute(3, 0, 1, 2)

    fast_stride = 2
    fast_idx = np.arange(32) * fast_stride
    fast = torch.from_numpy(raw_clip[fast_idx]).permute(3, 0, 1, 2)

    return [slow, fast], label
```

GitHub Link:

AbdollaZ/seniorproject_signlanguage: This is a repository for hosting CSCI 409 Senior Project…

This is a repository for hosting CSCI 409 Senior Project documentation and files. Project topic is Sign Language…

GitHub

THANK YOU